

FP3.1 Unit description

Further matrix algebra; vectors, hyperbolic functions; differentiation; integration, further coordinate systems

FP3.3 Assessment information**Prerequisites**

A knowledge of the specifications for C1, C2, C3, C4 and FP1, their prerequisites, preambles and associated formulae is assumed and may be tested.

Examination

The examination will consist of one 1½ hour paper. It will contain about eight questions with varying mark allocations per question which will be stated on the paper. All questions may be attempted.

Questions will be set in SI units and other units in common usage.

Calculators

Students are expected to have available a calculator with at least the following keys: $+$, $-$, $\times$, $\div$, π , x^2 , $\sqrt{x}$, $\frac{1}{x}$, x^y , $\ln x$, e^x , $x!$, sine, cosine and tangent and their inverses in degrees and decimals of a degree, and in radians; memory. Calculators with a facility for symbolic algebra, differentiation and/or integration are not permitted.

1 Hyperbolic functions

What students need to learn:

Definition of the six hyperbolic functions in terms of exponentials. Graphs and properties of the hyperbolic functions.

For example, $\cosh x = \frac{1}{2}(e^x + e^{-x})$,

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}.$$

Students should be able to derive and use simple identities such as $\cosh^2 x - \sinh^2 x = 1$ and $\cosh^2 x + \sinh^2 x = \cosh 2x$ and to solve equations such as $a \cosh x + b \sinh x = c$.

Inverse hyperbolic functions, their graphs, properties and logarithmic equivalents.

Eg $\operatorname{arsinh} x = \ln[x + \sqrt{1 + x^2}]$.

Students may be required to prove this and similar results.

2 Further Coordinate Systems

What students need to learn:

Cartesian and parametric equations for the ellipse and hyperbola.

Extension of work from FP1.

Students should be familiar with the equations:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 ; x = a \cos t, y = b \sin t.$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 ; x = a \sec t,$$

$$y = b \tan t, x = a \cosh t, y = b \sinh t.$$

The focus-directrix properties of the ellipse and hyperbola, including the eccentricity.

For example, students should know that, for the ellipse, $b^2 = a^2(1 - e^2)$, the foci are $(ae, 0)$ and $(-ae, 0)$ and the equations of the directrices are $x = +\frac{a}{e}$ and $x = -\frac{a}{e}$.

Tangents and normals to these curves.

The condition for $y = mx + c$ to be a tangent to these curves is expected to be known.

Simple loci problems.

3 Differentiation

What students need to learn:

Differentiation of hyperbolic functions and expressions involving them.

For example, $\tanh 3x$, $x \sinh^2 x$,
 $\frac{\cosh 2x}{\sqrt{x+1}}$.

Differentiation of inverse functions, including trigonometric and hyperbolic functions.

For example, $\arcsin x + x\sqrt{1-x^2}$,
 $\frac{1}{2} \operatorname{artanh} x^2$.

4 Integration

What students need to learn:

Integration of hyperbolic functions and expressions involving them.

Integration of inverse trigonometric and hyperbolic functions.

Integration using hyperbolic and trigonometric substitutions.

Use of substitution for integrals involving quadratic surds.

The derivation and use of simple reduction formulae.

The calculation of arc length and the area of a surface of revolution.

For example,

$$\int \operatorname{arsinh} x \, dx, \int \arctan x \, dx.$$

To include the integrals of $1/(a^2 + x^2)$, $1/\sqrt{(a^2 - x^2)}$, $1/\sqrt{(a^2 + x^2)}$, $1/\sqrt{(x^2 - a^2)}$.

In more complicated cases, substitutions will be given.

Students should be able to derive formulae such as

$$nI_n = (n-1)I_{n-2}, \quad n \geq 2,$$

$$\text{for } I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx,$$

$$I_{n+2} = \frac{2 \sin(n+1)x}{n+1} + I_n$$

$$\text{for } I_n = \int \frac{\sin nx}{\sin x} \, dx, \quad n > 0.$$

The equation of the curve may be given in cartesian or parametric form. Equations in polar form will not be set.

5 Vectors

What students need to learn:

The vector product $\mathbf{a} \times \mathbf{b}$ and the triple scalar product $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}$.

Use of vectors in problems involving points, lines and planes.

The equation of a line in the form $(\mathbf{r} - \mathbf{a}) \times \mathbf{b} = \mathbf{0}$.

The equation of a plane in the forms $\mathbf{r} \cdot \mathbf{n} = p$, $\mathbf{r} = \mathbf{a} + s\mathbf{b} + t\mathbf{c}$.

The interpretation of $|\mathbf{a} \times \mathbf{b}|$ as an area and $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}$ as a volume.

Students may be required to use equivalent cartesian forms also.

Applications to include

- (i) distance from a point to a plane,
- (ii) line of intersection of two planes,
- (iii) shortest distance between two skew lines.

Students may be required to use equivalent cartesian forms also.

6 Further Matrix Algebra

What students need to learn:

Linear transformations of column vectors in two and three dimensions and their matrix representation.

Combination of transformations. Products of matrices.

Transpose of a matrix.

Evaluation of 3×3 determinants.

Inverse of 3×3 matrices.

The inverse (when it exists) of a given transformation or combination of transformations.

Eigenvalues and eigenvectors of 2×2 and 3×3 matrices.

Reduction of symmetric matrices to diagonal form.

Extension of work from FP1 to 3 dimensions.

The transformation represented by $\mathbf{AB}$ is the transformation represented by $\mathbf{B}$ followed by the transformation represented by $\mathbf{A}$.

Use of the relation $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$.

Singular and non-singular matrices.

Use of the relation $(\mathbf{AB})^{-1} = \mathbf{B}^{-1} \mathbf{A}^{-1}$.

Normalised vectors may be required.

Students should be able to find an orthogonal matrix $\mathbf{P}$ such that $\mathbf{P}^T \mathbf{A} \mathbf{P}$ is diagonal.